Does Default Stay Local?
State Default and City Credit in the 1840s

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Abstract

In 1843 the federal government refused to take over the debts of American states that had stopped paying their creditors, leaving each state to stand on its own credit. This paper asks whether a state's default also raised borrowing costs for the cities inside it. Using weekly bond prices from the 1830s to the 1850s, I compare the yields on city bonds with the yields on their own state's bonds. When Pennsylvania defaulted, its borrowing costs rose but those of Philadelphia and Pittsburgh did not: the gap between Pennsylvania and Philadelphia widened by about 1.2 percentage points, all of it from the state's side, and Pittsburgh traded well below the state as well. In Ohio, which never defaulted, Cincinnati and the state traded at nearly the same yield. At the state level, Pennsylvania's premium over Ohio was about 2 percentage points, much smaller than a standard calculation suggests. The results show that investors in the 1840s judged each government on its own finances rather than on where it was located.

JEL codes: H74, H77, N21, G12
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1 Introduction

When a government stops paying its debts, investors charge it more to borrow. A harder question is how far that punishment spreads. In a country made up of layers of government—a national government, states inside it, and cities inside the states—a default at one layer could easily raise borrowing costs at the others. Investors might reason that trouble in a state signals trouble for everything inside it. Or they might look past the shared name and territory and judge each government on its own taxes and its own debts.

This paper studies that question using the American state debt crisis of the 1840s. In the 1830s, states borrowed heavily, much of it from British and Dutch investors, to build canals and railroads. After the financial panic of 1837, eight states and the Territory of Florida stopped paying their creditors (Wallis et al. 2004). In early 1843 the federal government made clear that it would not take over their debts, breaking with the precedent Alexander Hamilton had set in 1790. From then on, each state stood on its own credit. That episode has long been studied as an early test of what default costs a government (English 1996), of how individual states managed their budgets (Hall and Sargent 2025, 2026), and of how a federal system can be made to work without bailouts (Sargent 2012). This paper asks what it meant one level down: when a state defaulted, did its cities pay for it too?

To answer, this paper builds yields from raw weekly bond prices recorded in nineteenth-century New York and Philadelphia, and compares each city's bonds with its own state's. The clearest case is Pennsylvania, which defaulted in August 1842, and its two largest cities, Philadelphia and Pittsburgh, which did not. Cincinnati, whose state of Ohio never defaulted, provides a comparison where the state was not in trouble.

The answer is that the state's default did not reach its cities. After the federal government's refusal, the gap between Pennsylvania's yield and Philadelphia's widened by about 1.2 percentage points. All of that came from the state's yield rising; Philadelphia's yield did not rise at all. Pittsburgh, the state's second city, traded about 2.5 percentage points below Pennsylvania after the refusal, close to Philadelphia's 2.7. Cincinnati and Ohio, by contrast, traded at almost exactly the same yield. Investors drew a clear line between a defaulting state and the separately financed cities inside it, and drew it only when the state was in trouble.

That result sits on top of a more modest finding about the states themselves. Pennsylvania's premium over Ohio after the refusal is about 2 percentage points once yields are measured in a way that does not assume a defaulting state will repay on time; the usual measure suggests more than 13. Alabama, which never defaulted, still paid about 2 points more than Ohio at its peak. And within Indiana's restructured debt, bonds paid second yielded about 30 points more than bonds paid first. Across all of these comparisons, investors priced each claim by who owed it and what stood behind it.

Section 2 describes the borrowing, the defaults, and the federal refusal. Section 3 explains how the data were built and measured. Section 4 sets out the state-level evidence, and Section 5 presents the city-level results. Section 6 considers what they mean, including for today's debate over shared

debt in Europe, and Section 7 concludes.

2 Background

2.1 Borrowing to Build

In the 1830s, American states borrowed heavily to pay for canals, railroads, and banks. Much of this money came from investors in Britain and the Netherlands, who bought state bonds through merchant banks in London and Amsterdam (Kim and Wallis 2005). The logic was simple: a canal or railroad would raise trade, trade would raise tax revenue and tolls, and that revenue would repay the bonds. Pennsylvania borrowed to build the Main Line of Public Works, a chain of canals and railroads meant to link Philadelphia to Pittsburgh. Indiana borrowed for the Wabash and Erie Canal and a larger program of roads and canals. Ohio financed its own canal system the same way.

2.2 Panic and Default

The plan depended on the economy continuing to grow, and it did not. A financial panic in 1837, followed by a second downturn in 1839, cut into trade, land sales, and tax collections just as many projects were still unfinished and earning little or nothing. States found themselves owing interest on debts for works that were not yet paying for themselves.

Between 1841 and 1843, eight states and the Territory of Florida stopped paying their creditors: Arkansas, Illinois, Indiana, Louisiana, Maryland, Michigan, Mississippi, and Pennsylvania (Wallis et al. 2004). Their responses differed. Some, including Pennsylvania, suspended interest for a few years and then resumed paying, with back interest. Others, including Mississippi and Florida, refused to pay part or all of what they owed. Indiana settled with its creditors in 1846–47 by handing over its largest canal in exchange for forgiveness of about half its debt.

Not every state that looked risky defaulted. Alabama faced serious trouble in its state-owned banks but raised taxes and sold bank assets to keep paying. New York, which had also borrowed heavily for canals, kept paying throughout. Ohio came through the crisis without a default. These differences are what make the comparisons in Section 4 possible.

2.3 The Federal Government Says No

The question hanging over all of this was whether the federal government would step in. There was a precedent that suggested it might. In 1790, under a plan designed by Alexander Hamilton, the new federal government took over the debts the states had run up fighting the Revolutionary War. Foreign investors in the 1830s had reason to think that if the states got into trouble, Washington would do the same again. Some bankers and investors expected or hoped for exactly that, though others doubted it.

Washington did not step in. In the congressional session that ended in early 1843, members of both the House and the Senate took up the question of assuming the states' debts, and the answer

that emerged was no. Following the historical accounts of this episode, this paper dates that signal to February 11, 1843.[1] The message was repeated to European bankers over the following months. In the spring of 1843, bankers led by the Amsterdam house of Hope & Co. pressed the American consul Christopher Hughes about the state debts. Writing back that summer, he told them that the United States did not "consent, to be held, in any wise, or to any extent, responsible for any default, actual or eventual" (Thomson 2024).

From that point, a state's bonds were only as good as the state itself. That is the setting this paper studies. If investors believed the federal government's refusal, a defaulting state should have kept paying more to borrow than a state that never defaulted, which Section 4 tests. The paper's main question, taken up in Section 5, goes one level down: if a state had to stand on its own credit, did its cities have to stand on the state's, or on their own?

3 Data and Method

This paper works from raw nineteenth-century price records rather than from published yield series, so most of the work behind it was computational: turning scattered weekly price quotes into clean, comparable yield series, and then deciding how much confidence those series can support. This section describes that process in order—where the data come from, how they were assembled, how prices were converted to yields, and how the comparisons in Sections 4 and 5 were measured.

3.1 Sources

All prices come from the Early U.S. Securities Prices database hosted by EH.net (Sylla et al. n.d.), which transcribes bond and stock quotations from nineteenth-century newspapers and price lists. Two of its city files are used: New York and Philadelphia. Each file records, for each date, the quoted price of each security, identified only by a short code. What those codes mean is recorded separately in the database's codebook, which lists 2,152 securities by code, name, type, interest rate, and maturity date. State bonds carry codes beginning with "S-" and city bonds codes beginning with "C-".

3.2 Building the Dataset

The raw files were not built for analysis, and assembling them into one usable table was the first real task. All processing was done in Python as a chain of scripts: each one reads a file, performs one step, and saves its result for the next script to use.

[1] The date follows the secondary literature on the episode; see McGrane (1935, 21–26) for the 1843 debate over assuming the state debts. Speeches on assuming the state debts, by Representative William Gwin of Mississippi in the House and Senator George McDuffie of South Carolina, appear in the Appendix to the Congressional Globe, 27th Congress, 3rd session. Gwin also appears in a related debate over Mississippi's bonds in the main volume (pp. 283–284). None of these pages carries a date stamp, so the exact day rests on the secondary accounts. Because the analysis leaves out the seven weeks after this date, a difference of a few days would not change any result.

Three problems had to be solved along the way. First, the price files have no built-in labels, so every price had to be matched against the codebook by its code before it meant anything. Second, the New York file splits state bonds across two separate sheets. An early version of the parser combined the two sheets without recording which sheet each row came from, and as a result several Ohio, Alabama, and Indiana bonds were mislabeled as New York bonds. The fix was to label every row with the sheet it came from before combining them, and every later file was checked for the same problem. Third, city bonds turned out to be on a sheet of their own ("Municipal Debt" in Philadelphia's file), which the original script never read; the city comparisons in Section 5 required reading that sheet separately.

The finished state-level table contains 3,711 observations, each one a single bond's price on a single date, across the five states studied. Each row records the bond, the date, the quoted price, the computed yield, which yield formula was used, and flags for any rule that excluded or modified the observation.

3.3 From Prices to Yields

Every comparison in this paper is made in terms of *yield*: the yearly return a buyer earns on a bond at its current price. Yield is the standard way to read a bond market. When investors see more risk in a borrower, they pay less for its bonds, and a lower price means a higher yield. So a higher yield means investors were charging that government more to borrow. Differences in yield are measured in *percentage points*: a bond yielding 8 percent pays 2 percentage points more than one yielding 6 percent.

Prices alone cannot be compared across bonds, because bonds paid different interest rates and came due at different times. Each bond promised a fixed yearly interest payment, called its *coupon*, and repayment of its $100 face value on a set date, called its *maturity*. Two measures of yield are used.

The simpler is *current yield*: the yearly interest payment divided by the price. For a bond with yearly coupon C (in dollars per $100 of face value) trading at price P,

$$y_{\text{current}} = \frac{C}{P} \times 100. \tag{1}$$

A 5 percent bond trading at $80 has a current yield of $5/80 = 6.25$ percent.

The second is *yield to maturity*, which also accounts for the gain or loss a buyer makes when the bond is repaid at its face value of $100. With T years left until maturity, the paper uses the standard approximation

$$y_{\text{maturity}} \approx \frac{C + (100 - P)/T}{(100 + P)/2} \times 100. \tag{2}$$

The numerator is the yearly income from the bond: its interest plus the eventual repayment gain spread evenly over the years remaining. The denominator is the average of the purchase price and the repayment value.

Yield to maturity is the more complete measure, but it has two weak points, and handling them turned out to matter a great deal.

Bonds close to or past maturity. Because T sits in the denominator of the repayment term, equation (2) becomes unstable as a bond nears its due date: small price changes produce large swings in the computed yield. One New York bond, quoted 23 days before maturity, produced a yield of -8.8 percent, which has no economic meaning. Some bonds also kept trading after their stated maturity date had passed, where T is negative and the formula is undefined. Rather than patch the formula, observations within twelve months of maturity and all observations past maturity are excluded from the main yield series.

Bonds in default. Equation (2) assumes the bond will be repaid in full and on time. For a state that has stopped paying its creditors, that assumption is false, and the formula can produce badly inflated yields. The clearest case was one Pennsylvania bond whose yield to maturity averaged about 30 percent across its entire post-default record, compared with about 10 percent on a current-yield basis. The rule adopted is simple: for any date on which a state was in default, current yield replaces yield to maturity, regardless of the bond's maturity. The default periods used are August 1842 to February 1845 for Pennsylvania and January 1841 onward (through the end of the data) for Indiana; Alabama and the other states never defaulted in this period. This single rule lowered Pennsylvania's estimated post-signal premium over Ohio from about 13 percentage points to about 2 (Section 4).

Two further cases use current yield for a different reason: Alabama's and Indiana's bonds, and the city bonds, have no maturity date in the codebook, so equation (2) cannot be computed for them at all.

3.4 Grouping States and Bonds

States are placed in three groups according to how they came through the crisis: *defaulted* (Pennsylvania, Indiana), *risky but survived* (Alabama, New York), and *safe* (Ohio). Alabama was originally grouped with the defaulters; historical sources show it raised taxes and sold state-bank assets to keep paying, and it was reclassified accordingly.

Within each state, bonds also differ in what backs them. Most were general obligations, repaid from the state's overall revenue. Some, however, were repaid only from a specific income stream such as canal tolls, which makes them a different kind of risk. The codebook's type field does not distinguish these—it records only whether a security is a stock or a bond—so the name of every Pennsylvania, Ohio, Alabama, Indiana, and New York entry (326 in total) was searched by script for words indicating a revenue pledge. This search showed that the three New York bonds first chosen for the analysis were all canal bonds, which would have compared a toll-backed bond against four states' general obligations. The main comparison was rebuilt using general-obligation bonds only. The canal bonds are reported separately, alongside Indiana's restructured canal bonds

(Sections 4.3 and 4.4).

3.5 Two Time Windows

Two different events could have moved bond prices, and they are kept apart throughout. The first is the financial panic that began in 1837 and the wave of defaults that followed it—an economic shock. The second is the federal government's refusal to take over the states' debts, dated here to February 11, 1843 (Section 2)—a policy signal. The analysis counts observations before February 11, 1843 as "before" the signal and observations from April 1, 1843 onward as "after," leaving the seven weeks in between for the news to reach investors, including those in Europe. Mixing the two windows would risk crediting the policy signal with price movements that were really caused by the panic, or the reverse.

3.6 Comparing Series and Measuring Uncertainty

Most states have several bonds quoted on any given date. A state's yield on a date is the average of the yields of all its usable bonds quoted that day, and a yearly figure is the average of those daily values over the calendar year. Sample sizes reported in this paper count distinct dates, not individual bond quotes.

For each city–state pair, the gap on a date t is

$$g_t = y_t^{\text{state}} - y_t^{\text{city}}, \tag{3}$$

computed only on dates when both a state bond and the city bond were quoted, so that every gap compares prices from the same day. The reported gap is the average of g_t over the window.

Measuring how precise that average is requires care. The usual formula for a standard error assumes each observation is independent of the others. Weekly bond prices are not: a bond that traded at \$80 last week is very likely to trade near \$80 this week. Treating these observations as independent would count what is really one slowly moving price as dozens of separate pieces of evidence, making the estimates look far more precise than they are. The paper instead uses the Newey–West estimator, which accounts for how closely nearby weeks move together. If $\hat{\gamma}_j$ measures how closely gaps j weeks apart move together (their covariance), the variance used for the average of n gaps is

$$\hat{\sigma}^2 = \frac{1}{n}\left[\hat{\gamma}_0 + 2\sum_{j=1}^{L}\left(1 - \frac{j}{L+1}\right)\hat{\gamma}_j\right], \tag{4}$$

where the weights shrink for weeks further apart and L, the number of weeks of correlation allowed for, is set by the standard rule $L = \lfloor 4\,(n/100)^{2/9} \rfloor$. The standard error is $\hat{\sigma}$, and 95 percent confidence intervals are the average gap plus or minus $1.96\,\hat{\sigma}$.

For Philadelphia, where there is enough data both before and after the signal, the change in the gap is the after-signal average minus the before-signal average. Because the two windows do

not overlap, its standard error combines the two: $\sqrt{\hat{\sigma}^2_{\text{before}} + \hat{\sigma}^2_{\text{after}}}$.

3.7 Data Checks and Reproducibility

Several checks were built into the scripts to catch problems before they reached a result.

- **Gaps are shown, never filled.** For every bond, a script measures the longest stretch between quotes. Long gaps—such as the 343-day hole in the main Philadelphia bond—are drawn as breaks in the charts rather than bridged with straight lines, and no values are filled in.
- **Different bonds are not spliced together.** Where one bond's record ends just as another's begins, joining them into one series is tempting. It was rejected wherever the two bonds paid different interest rates or had different maturities, since the join would create an artificial jump at the seam.
- **Thinly traded bonds are flagged.** Each bond's quotes per month and longest gap were tabulated to identify bonds quoted too rarely to trust on their own.
- **Suspicious prices are kept but flagged.** Isolated prices far out of line with a bond's own history are left in the data files, so that the underlying record is never altered, but are kept off the charts and out of the averages they would distort—for example, a single quote of $160 for a bond otherwise trading near its $100 face value (Sections 4.3 and 4.4).
- **Every result can be regenerated exactly.** Each script was rerun from the raw files and checked to produce exactly the same tables and charts.

3.8 Sample and Coverage

Table 1 lists the state bonds used in the main comparison and how many usable observations each has before and after the no-bailout signal. How far the data reach varies a lot by state. Only Ohio's and Alabama's bonds are quoted regularly into the 1850s; Pennsylvania's usable record ends in January 1845, and Indiana's and New York's in 1848. New York City and Brooklyn bonds were also examined as possible additions to the city comparisons, but none of the 63 candidate bonds has usable prices from before the signal, so they cannot show whether anything changed and are left out.

Table 1: State Bonds in the Main Comparison

State	Bond	Before signal	After signal	Last usable quote
Pennsylvania	S-2240	340	0	Aug 1839
	S-2250	433	0	Sep 1842
	S-2270	442	5	Dec 1843
	S-2330	—	93	Dec 1844
	S-2410	—	97	Jan 1845
Ohio	S-2100	464	21	Oct 1843
	S-2110	45	116	Dec 1853
	S-2080	75	423	Dec 1853
	S-2010	159	5	Jul 1848
Alabama	S-0030	18	140	Jun 1853
	S-0040	10	9	May 1851
Indiana	S-0510	3	187	Dec 1848
	S-0540	3	134	Sep 1848
New York	S-1650	17	117	Jan 1848

Notes: Counts are dates with a usable price, after the maturity exclusions described in the text. "Before" is before February 11, 1843; "after" is from April 1, 1843. *Source:* Author's calculations from the EH.Net Early U.S. Securities Prices database (Sylla et al. n.d.).

4 Setting the Stage: State Bond Yields

Before turning to cities, it helps to establish a simpler point: did the federal government's refusal to bail out the states show up in state bond prices at all? This section is setup rather than the main result. The paper's contribution is what happens at the city level (Section 5), but that question only matters if investors were treating defaulting states differently in the first place.

4.1 Pennsylvania: A Smaller Premium Than It First Appeared

Pennsylvania did pay more, though by less than a first look suggests. After the no-bailout signal, Pennsylvania's bonds yielded an average of 8.77 percent, about 2.1 percentage points more than Ohio's 6.69 percent. Measured the usual way, the gap looks much bigger, above 13 points, but that measure assumes a bond will be repaid in full and on schedule—an assumption that clearly did not hold while Pennsylvania was in default. Using a measure that does not depend on that assumption brings the gap down to about 2 points.[2] Pennsylvania still paid a premium, just a much smaller one.

4.2 Alabama: Priced as Risky Without Defaulting

Alabama tells a similar story from a different starting point. It never defaulted: it raised taxes and sold off state-bank assets to keep paying its creditors. Even so, investors charged it more than Ohio

[2]During each state's default period, yield-to-maturity is replaced with current yield (annual interest divided by price). See Section 3.

Table 2: Alabama vs. Ohio Bond Yields, Selected Years

Year	Alabama	Ohio	Gap (pp)
1843	8.23%	8.18%	+0.04
1844	6.24%	6.13%	+0.12
1845	7.11%	6.31%	+0.81
1847	8.26%	6.22%	+2.04
1848	8.12%	6.11%	+2.01
1853	5.45%	4.44%	+1.01

Notes: Yearly averages of each state's daily yield. pp = percentage points. *Source:* Author's calculations from the EH.Net Early U.S. Securities Prices database (Sylla et al. n.d.).

for most of a decade. Table 2 traces the gap: nearly zero in 1843, about 2 points by 1847–48, and still about 1 point in 1853. The table shows selected years to trace this path; the years in between follow the same pattern.

4.3 New York and Indiana

New York shows no premium over Ohio at all, but this should not be read as proof that investors saw New York as risk-free. The price record for its main bond begins only in July 1842, after New York's own worst stretch had passed, so the data may simply miss the period when New York looked shakiest. New York's canal bonds, which were repaid from canal tolls and so are kept out of the main comparison, have a longer record, and they point the same way: their average yield fell from 6.10 percent before the signal to 5.26 percent after it.[3]

Indiana, by contrast, stayed in default for the entire period covered here. Its 1846–47 settlement handed the Wabash and Erie Canal to creditors in exchange for forgiving about half the debt. That was a restructuring, not a return to normal payment.

4.4 Indiana's Restructured Bonds: Who Gets Paid First

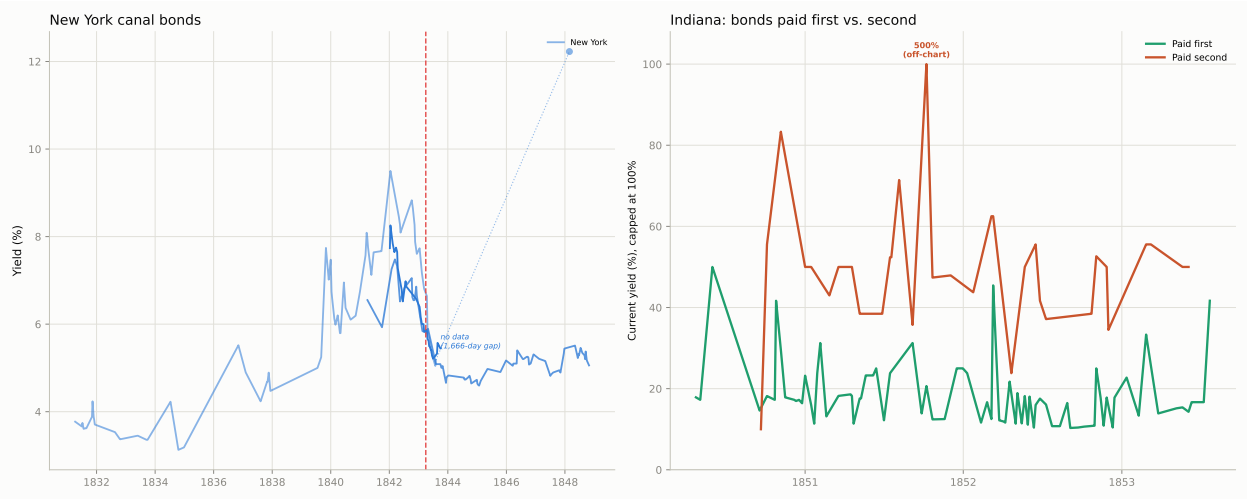
Indiana's settlement offers a different kind of test. Rather than comparing one government with another, it compares two kinds of bonds issued by the same state and repaid from the same canal. After the settlement, Indiana's canal debt traded in New York as two groups of bonds. Holders of the first group had first claim on the canal's toll income. Holders of the second group ranked behind them.[4]

Investors priced that difference sharply. Over the period when both groups were trading, from September 1850 to June 1853, the bonds paid second yielded about 30 percentage points more on

[3]Bonds S-1750, S-1820, and S-1950; 96 dates before the signal and 78 after. One quote of $160 for bond S-1820, almost certainly a recording error, is left out.

[4]The two groups traded as "preferred" (bonds S-0490 and S-0500) and "deferred" (bonds S-0480 and S-0506). These were market labels; the settlement laws themselves do not use them.

Figure 1: Canal Bond Yields: New York's Canal Bonds and Indiana's Restructured Bonds



Notes: Left panel: yields on New York's three canal bonds (S-1750, S-1820, S-1950), which were repaid from canal tolls; the dashed line marks April 1843, and the dotted segment marks a stretch with no quotes, ending at the next quote (marked with a dot). Right panel: current yields on Indiana's bonds paid first (S-0490, S-0500) and paid second (S-0480, S-0506) after the 1846–47 settlement, capped at 100 percent; one quote above the cap is labeled. *Source:* Author's calculations from the EH.Net Early U.S. Securities Prices database (Sylla et al. n.d.).

average than the bonds paid first: 48.4 percent against 18.0 percent.[5] The bonds paid first typically traded between $20 and $42 per $100 of face value; the most junior bond typically traded near $9. This is the largest gap anywhere in the data. Its exact size is sensitive to a single sale of the most junior bond at $1, which implies a yield of 500 percent, but the gap does not depend on it: the bonds paid second yielded far more than the bonds paid first in every one of the 25 months in which both were quoted (Figure 1).

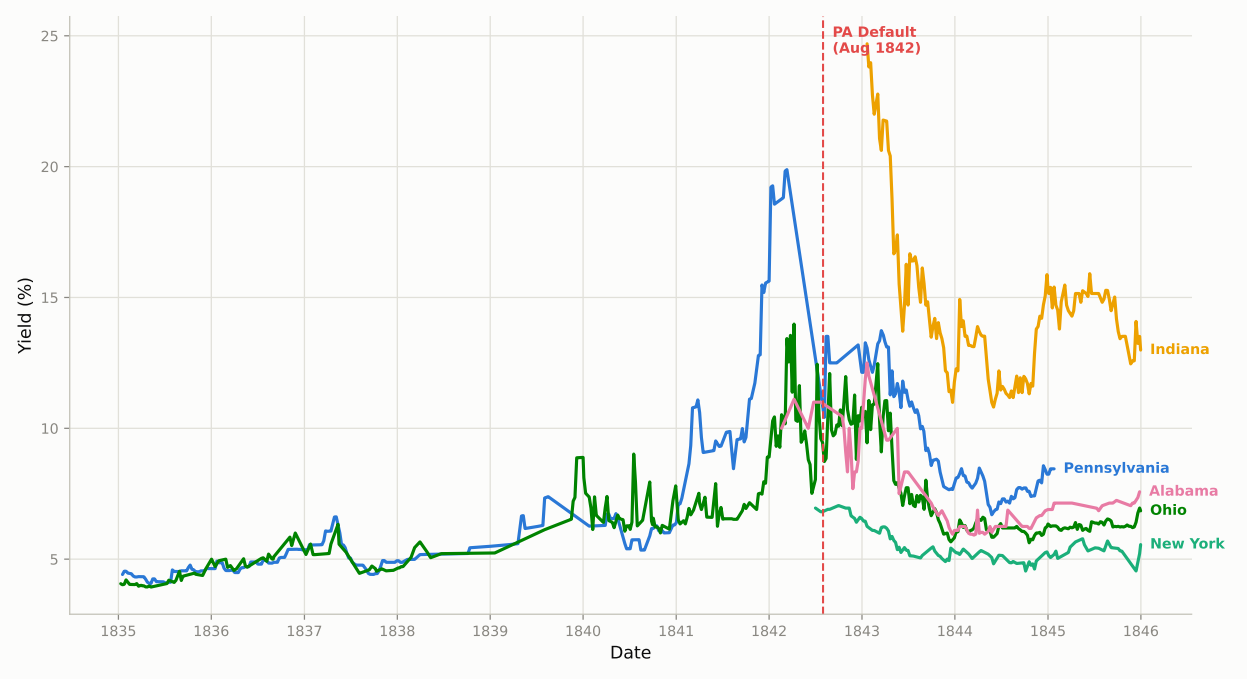
One detail remains unknown. The sources found do not say exactly how a shortfall in canal income was divided between the two groups—whether the first group had to be paid in full before the second received anything, or whether both took a share of the loss. The prices suggest the second: even the bonds paid first traded far below $100, so investors expected losses on both. This affects how the arrangement is described, not the size of the gap.

4.5 All Five States Together

Figures 2 and 3 show all five states, first around the 1837–42 financial panic and then after the no-bailout signal, with each state's line ending where its usable data end.

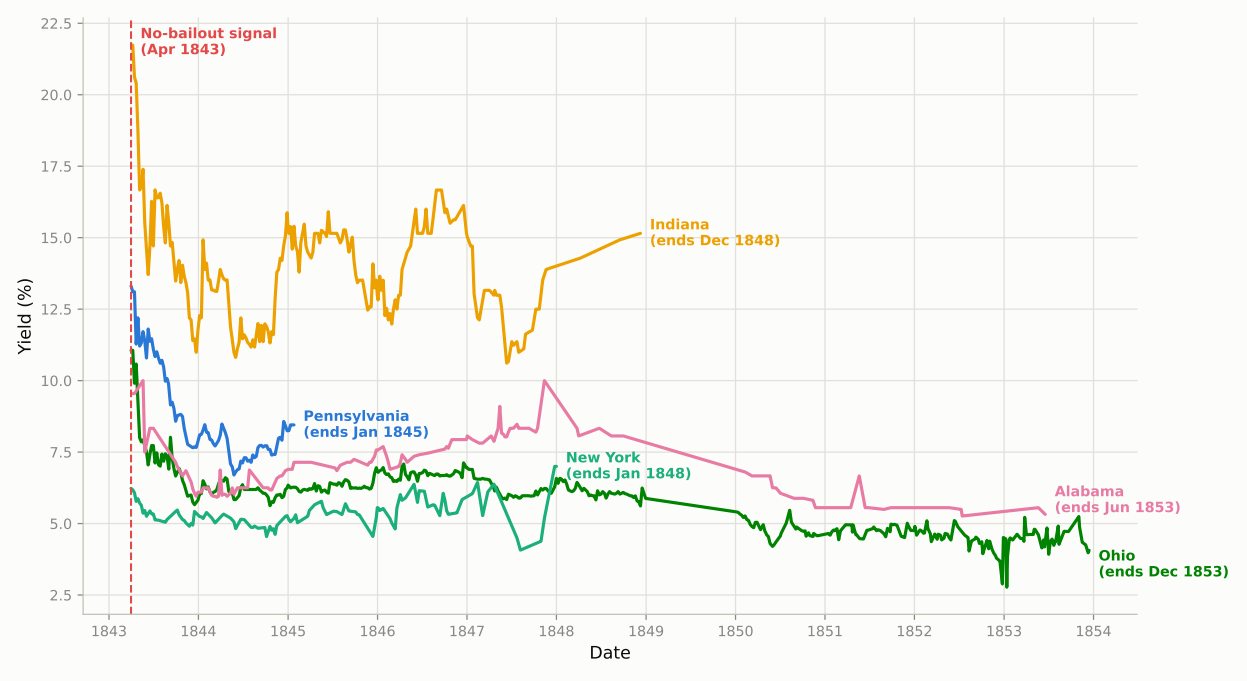
[5]Both groups have no maturity date and are measured by current yield. The averages are taken over 94 quotes for the bonds paid first and 41 for the bonds paid second. Two flagged quotes are left out: a first quote of $50 for bond S-0480, far above its later prices, and a single sale of bond S-0506 at $1. Counting the $1 sale raises the gap to 41.2 points; counting both quotes gives 40.0.

Figure 2: State Bond Yields Around the Panic, 1835–1845



Notes: Each line is the average yield across a state's usable bonds on each date. Yield to maturity is used for Pennsylvania, Ohio, and New York, except during a state's default, when current yield is used; Alabama and Indiana use current yield throughout. The dashed line marks Pennsylvania's default in August 1842. New York's bond is quoted only from July 1842. *Source:* Author's calculations from the EH.Net Early U.S. Securities Prices database (Sylla et al. n.d.).

Figure 3: State Bond Yields After the No-Bailout Signal

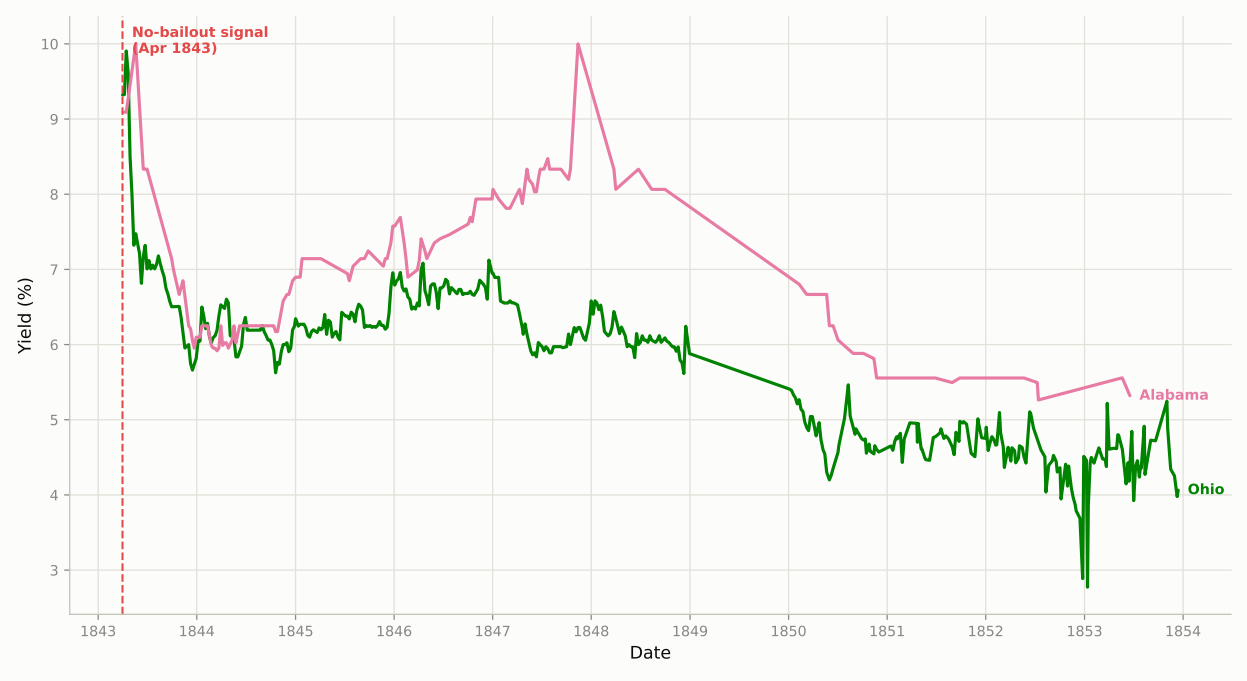


Notes: Measured as in Figure 2. Each line ends where that state's usable data end. The dashed line marks April 1843, the start of the period after the signal. *Source:* Author's calculations from the EH.Net Early U.S. Securities Prices database (Sylla et al. n.d.).

4.6 How Far the Data Reach

Only Ohio and Alabama have enough price data to follow into the 1850s (Figure 4). Pennsylvania's usable data end in January 1845, less than two years after the signal, even after searching all 34 of its bonds in the source records. Indiana's and New York's end in late 1848. So the claim that a premium lasted is much better supported for Alabama than for Pennsylvania, simply because Alabama's data run longer.

Figure 4: Ohio and Alabama Bond Yields Through the 1850s



Notes: Ohio's and Alabama's are the only state series quoted regularly into the 1850s. Straight segments span periods without quotes. The dashed line marks April 1843. *Source:* Author's calculations from the EH.Net Early U.S. Securities Prices database (Sylla et al. n.d.).

5 The Central Result: Cities and Their States

This section asks the paper's main question: when a state defaulted, did investors also charge more to lend to its cities? Each city's bonds are compared with its own state's bonds, first for Philadelphia and Pittsburgh, whose state defaulted, and then for Cincinnati, whose state did not.

5.1 Philadelphia: The State's Default Did Not Reach the City

Pennsylvania defaulted on its debt in August 1842. Philadelphia, the state's largest city, did not. The two were separate governments, with their own power to borrow, their own sources of revenue, and no responsibility for each other's debts. Pennsylvania had borrowed to build the Main Line of Public Works—a canal and railroad system that cost more than $18 million—along with other canals and investments in bank stock. Philadelphia had borrowed for its waterworks, gas works, wharves, and public buildings, and repaid that debt from city property taxes and utility fees rather than the canal tolls and state taxes that collapsed along with Pennsylvania's finances.

Given this separation, did investors treat Philadelphia's debt as separate from Pennsylvania's, or did the state's default drag the city's borrowing costs up too? The no-bailout signal is dated to February 11, 1843 (Section 2). Because news took time to reach investors, including those in Europe, the analysis counts April 1, 1843 as the start of the "after" period and sets aside the seven weeks in between.

Before the signal, Philadelphia already borrowed more cheaply than Pennsylvania: on dates when both bonds were quoted, the city's yield was 1.49 percentage points below the state's. After the signal the gap grew. From February to December 1844, Pennsylvania's bonds yielded 7.56 percent and Philadelphia's 4.88 percent, a gap of 2.68 points.[6]

Because Philadelphia was already cheaper beforehand, what matters is whether the gap changed, not how big it was. It rose from 1.49 points before the signal to 2.68 after, an increase of 1.19 points. The 95 percent confidence interval for that increase runs from 0.36 to 2.01 points, so it does not include zero.[7]

The way the gap grew is the key point. Pennsylvania's yield rose by about 0.9 points while Philadelphia's fell slightly. If the state's trouble had spread to the city, Philadelphia's yield should have risen too, even if by less. It moved the other way. The city was not pulled partway up with the state; it simply did not move while the state did.

Figure 5 shows this directly. Pennsylvania's yield climbs toward 20 percent in early 1842, in the run-up to default, and stays well above its old level afterward. Philadelphia's stays close to 5 percent throughout.

5.2 Filling the Gap in Philadelphia's Record

One weakness in the Philadelphia comparison is a 343-day hole in the main Philadelphia bond's price record, from February 1843 to February 1844, which covers most of the months right after the signal. A second Philadelphia bond fills that hole and shows the same pattern: the gap during those months was 5.95 points from April to August 1843, and 6.28 points over the full period.[8] If anything, the gap was largest right after the signal. Even at the moment of greatest stress, Philadelphia's credit was not dragged toward Pennsylvania's.

5.3 Pittsburgh: The Same Gap in a Second City

Pittsburgh, like Philadelphia, was a separate city government with its own borrowing power and its own revenue. After the signal (April 1843 to January 1845, where Pennsylvania's data end), Pennsylvania's bonds yielded 8.76 percent and Pittsburgh's 6.11 percent, a gap of 2.65 points—almost exactly Philadelphia's 2.68. Using only dates when both were quoted, the gap is 2.48 points, with a 95 percent confidence interval of 1.95 to 3.01.[9]

Pittsburgh has only five price quotes from before the signal, too few to measure how the gap changed. The evidence here is therefore the size of the gap afterward, not a change in it. Still, the result shows that the Philadelphia finding is not a quirk of one bond or one city: the same gap, at

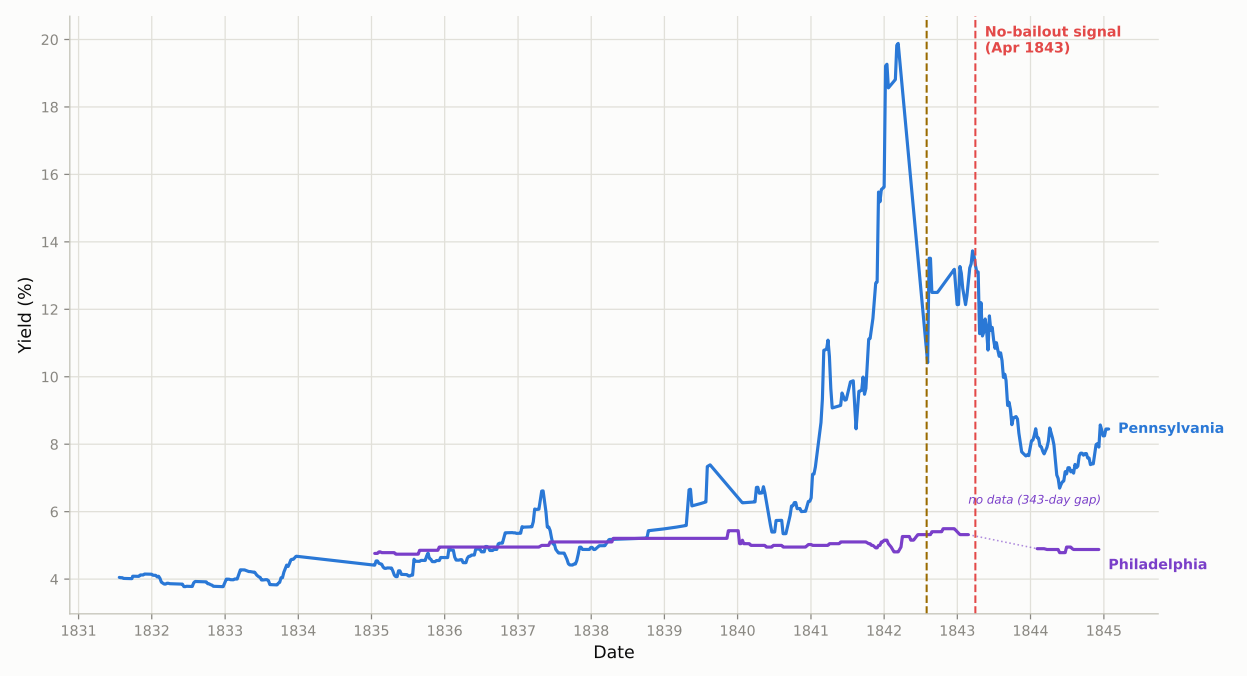
[6] City series: bond C-1100, "Philadelphia 5s, redeemable 1846." Both bonds were quoted on 328 dates before the signal and 41 dates between February and December 1844.

[7] Weekly bond prices are not independent of one another—each week's price tends to be close to the last—so ordinary standard errors would make the estimates look more precise than they are. All confidence intervals in this section use Newey–West standard errors, which correct for this.

[8] Bond C-1260. 19 dates for April–August 1843; 24 for the full gap period.

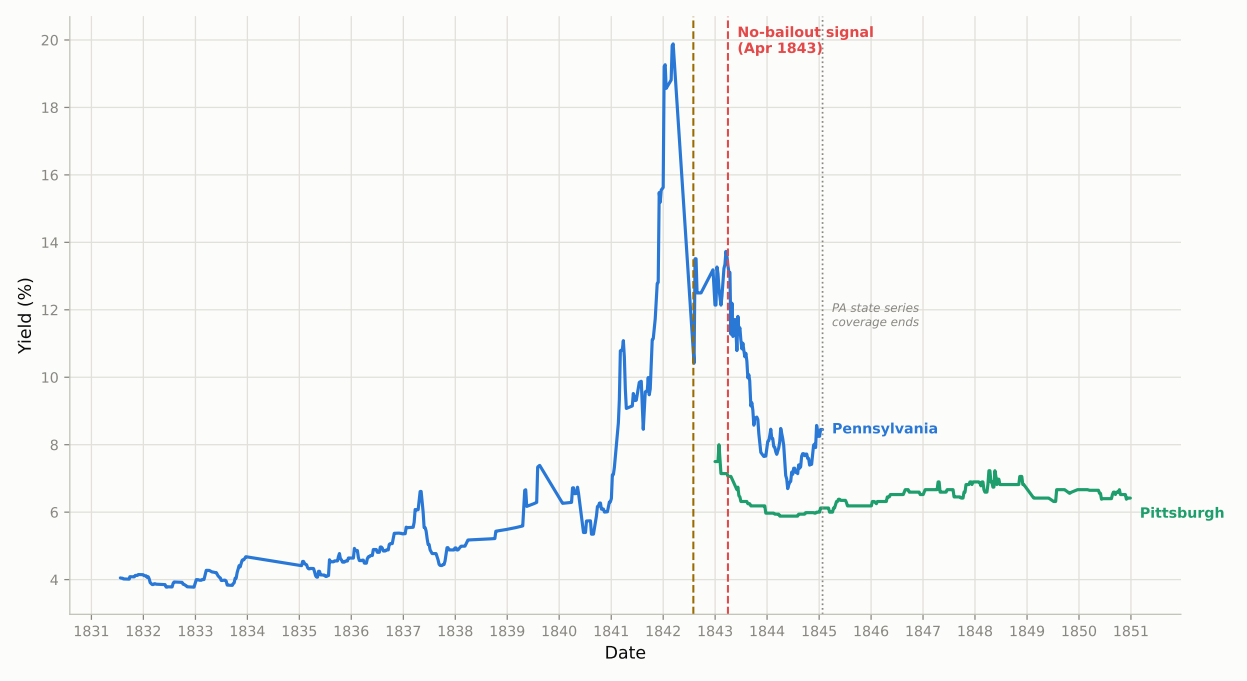
[9] City series: bond C-1394, "Pittsburgh 6s." 88 dates for the state and 83 for the city, all 83 of them matched.

Figure 5: Philadelphia and Pennsylvania Bond Yields, 1831–1845



Notes: Current yields on Pennsylvania's state bonds and on Philadelphia's bond C-1100. The two dashed lines mark Pennsylvania's default (August 1842) and the start of the period after the signal (April 1843). The dotted segment marks the 343-day gap in the Philadelphia bond's record, which is not filled in. *Source:* Author's calculations from the EH.Net Early U.S. Securities Prices database (Sylla et al. n.d.).

Figure 6: Pittsburgh and Pennsylvania Bond Yields



Notes: Current yields on Pennsylvania's state bonds and on Pittsburgh's bond C-1394. The dashed lines mark Pennsylvania's default (August 1842) and April 1843; the dotted line marks where Pennsylvania's usable data end (January 1845). *Source:* Author's calculations from the EH.Net Early U.S. Securities Prices database (Sylla et al. n.d.).

nearly the same size, appears in a second Pennsylvania city.

5.4 Cincinnati: No Gap When the State Did Not Default

Philadelphia and Pittsburgh show that a defaulting state's cities were not punished along with it. Cincinnati asks the reverse question. Ohio never defaulted. If the gap in Pennsylvania came from investors separating city from state when the state was in trouble—rather than cities always trading at a different level from their states—then Cincinnati and Ohio should trade at about the same yield.

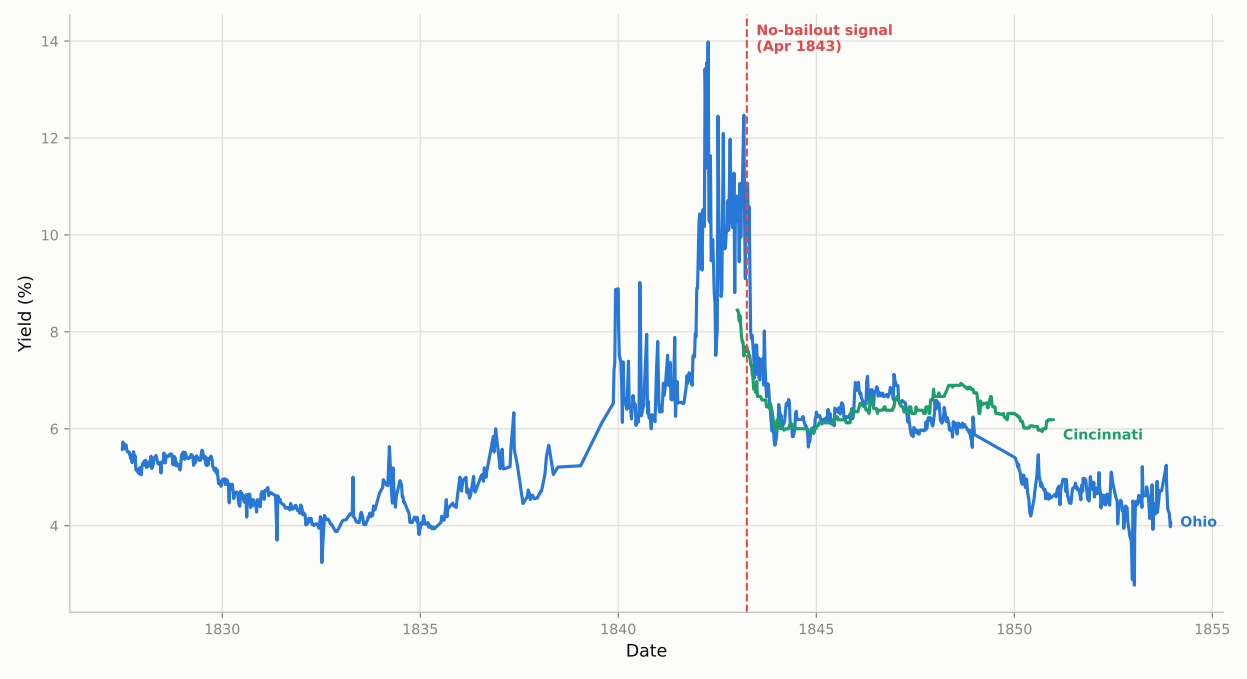
They do. After the signal, Ohio's bonds averaged 6.25 percent and Cincinnati's 6.36 percent. On dates when both were quoted, Cincinnati yielded 0.08 points more than Ohio, with a 95 percent confidence interval of −0.26 to 0.09 points, which includes zero.[10] In other words, there is no real gap to explain.

This tells us the Pennsylvania result is not simply how city and state bonds always relate. The gap opened when the state was in trouble and is absent when it was not.

Two cautions apply. First, as with Pittsburgh, there are only a handful of quotes from before the signal, so this comparison also rests on the period afterward. Second, the two sides are measured

[10]City series: bond C-0436, "Cincinnati 6s." April 1843–December 1850; 322 dates for the state and 339 for the city, 255 of them matched.

Figure 7: Cincinnati and Ohio Bond Yields



Notes: Ohio's state bonds, measured as in Figure 2, and Cincinnati's bond C-0436, measured by current yield. The dashed line marks April 1843. *Source:* Author's calculations from the EH.Net Early U.S. Securities Prices database (Sylla et al. n.d.).

slightly differently: Ohio's yields use the standard yield-to-maturity formula, while Cincinnati's bond has no recorded maturity date and uses current yield (annual interest divided by price). The difference between the two measures could not plausibly create or hide a gap of this size, but the comparison is not perfectly like-for-like.

5.5 What the Three Cases Show

The three cases point the same way. Pennsylvania's default raised the state's own borrowing costs but left its two largest cities' costs largely untouched. In Philadelphia, where there is enough data before and after the signal, the gap between city and state widened by 1.19 points, and all of that came from Pennsylvania's yield rising while Philadelphia's held steady or fell. After the signal, Pittsburgh traded about as far below the state as Philadelphia did, and a second Philadelphia bond shows the gap held even in the months right after the signal. Cincinnati, whose state never defaulted, shows no gap at all. Table 3 collects these results.

Investors, in other words, did not lump the city in with the state. They charged a defaulting government more without charging more to a separate, independently financed city that shared its name and its territory.

Table 3: Gap Between State and City Bond Yields

Pair and period	Dates	Gap (pp)	Std. error	95% interval
Philadelphia/PA, before signal	328	+1.49	0.404	[+0.70, +2.29]
Philadelphia/PA, after signal	41	+2.68	0.112	[+2.46, +2.90]
Philadelphia/PA, change	—	+1.19	0.420	[+0.36, +2.01]
Pittsburgh/PA, after signal	83	+2.48	0.269	[+1.95, +3.01]
Cincinnati/Ohio, after signal	255	−0.08	0.090	[−0.26, +0.09]

Notes: Gap = state yield minus city yield, in percentage points, using only dates when both bonds were quoted. Standard errors are Newey–West, which allow for correlation between nearby weeks. *Source:* Author's calculations from the EH.Net Early U.S. Securities Prices database (Sylla et al. n.d.).

5.6 Limits of the Evidence

The evidence has some clear limits. Only Philadelphia has enough data before the signal to measure a change in the gap; for Pittsburgh and Cincinnati the evidence is the gap afterward. This is three comparisons, not a large sample: Philadelphia is tested with two different bonds and Pittsburgh with one. New York City and Brooklyn would be natural additions, but neither has usable price data from before the signal, so a comparison for them could not tell us whether anything changed. Whether the same pattern holds for cities in other states that defaulted, such as Maryland or Louisiana, remains untested.

6 Discussion

6.1 Investors Followed the Legal Lines

The central result is that investors kept a defaulting state and its cities apart. When Pennsylvania stopped paying, its own borrowing costs rose, but Philadelphia's and Pittsburgh's did not. This is not the only outcome one might have expected. The cities sat inside the state and shared its economy, and Philadelphia was its commercial center. An investor worried about "Pennsylvania" in general might reasonably have marked down anything issued there. The prices show that investors did not.

What they appear to have done instead is follow the legal lines. A city was a separate government, with its own taxes, its own debts, and no duty to pay the state's. Philadelphia repaid its bonds from property taxes and utility fees that the state's collapse did not touch. Investors priced the city on those facts rather than on its address. Cincinnati, whose state never defaulted, points the same way: there, city and state traded at about the same yield. The gap between a city and its state opened only when the state itself was in trouble.

The rest of the evidence in this paper tells a consistent story about how carefully these investors read the fine print. Alabama never defaulted but was still charged more than Ohio for most of a decade, so investors were pricing risk, not just missed payments. And within a single state's debt, Indiana's bonds that were paid second yielded about 30 percentage points more than those paid

first (Section 4.4). In each case, the price depended on who owed the money and what stood behind it.

6.2 What the Result Does Not Show

Two cautions matter for how far this reaches. First, the evidence comes from three city–state pairs and one period, and only Philadelphia has enough data to measure how the gap changed across the episode. The pattern is consistent, but it is not a large sample. Second, a city insulated from its state's default is not a city with no connection to it. These cities did not default, but that is partly because their finances happened to rest on revenue the state's crisis did not reach. A city that depended on state payments, or that had guaranteed state projects, might well have been dragged down with it. The finding is that investors could tell the difference, not that the difference always existed.

6.3 A Modern Parallel

The same question arises today wherever governments are nested inside one another. The European Union is currently debating whether to issue joint "blue bonds" backed by member states together, and a central worry is whether doing so would weaken each country's incentive to manage its own debt (Lustig 2026). The United States in the 1840s offers one piece of evidence on how such a system can work. The federal government refused to stand behind the states, much as the EU's rules forbid it from standing behind its members. Pennsylvania's position relative to Washington is roughly a member state's position relative to the Union. This paper's result adds a second layer to that comparison: one level further down, Philadelphia's position relative to Pennsylvania. In the 1840s, investors drew the line at each level separately, charging each government for its own risk. Whether modern investors would draw those lines as cleanly is a separate question, but the historical record shows it can be done.

7 Conclusion

In early 1843, the federal government made clear that it would not take over the debts of states that had stopped paying. This paper asks what that meant one level down. If a state had to stand on its own credit, did its cities suffer along with it?

In the cases tested here, they did not. When Pennsylvania defaulted, its own borrowing costs rose, but Philadelphia's and Pittsburgh's did not. In Philadelphia, where the data allow a before-and-after comparison, the gap between the state's yield and the city's widened by about 1.2 percentage points, and all of that widening came from the state's yield rising while the city's held steady. After the signal, Pittsburgh traded about as far below the state as Philadelphia did. Cincinnati, whose state never defaulted, shows no gap at all. Investors treated each city as the separate government it was, with its own taxes and its own debts, and priced it accordingly.

The state-level evidence that sets up this question is more modest than it first appears. Once yields are measured in a way that does not assume a defaulting state will repay on schedule, Pennsylvania's premium over Ohio after the signal is about 2 percentage points, not 13. Alabama, which never defaulted, paid a premium of similar size at its peak and still paid about 1 point more than Ohio a decade later. Investors charged for risk, not only for missed payments.

Some questions remain open. The evidence rests on three city–state pairs, and only Philadelphia has enough data from before the signal to measure how the gap changed. New York City and Brooklyn could not be tested, and only Ohio and Alabama can be followed reliably into the 1850s. How Indiana's restructured canal bonds divided any shortfall between the bonds paid first and those paid later is not recorded in the sources found, though this does not change the size of the gap between them. The most natural next step is to test cities in other states that defaulted. The same database has price records for Baltimore and New Orleans, whose states, Maryland and Louisiana, both defaulted in this period.

The broader lesson is that investors in the 1840s drew credit lines carefully. When the federal government refused to stand behind the states, markets did not respond by treating everything inside a troubled state as troubled. They followed the legal boundaries between governments and charged each for its own risk.

Acknowledgments

The research question was proposed by the author, and the project was mentored by Thomas J. Sargent and George Hall, who advised throughout on what to focus on and what to set aside—for example, extending the analysis into the 1850s, separating the effect of the 1837 panic from that of the no-bailout signal, checking how each bond was backed and whether it was actively traded, and making the relationship between cities and their states the center of the paper. The author designed the analysis, directed each stage of the work, and verified all results against the source data before they were included in the paper. Analysis scripts were written by the author, with Anthropic's Claude Code used to help debug them. Where standard methods have known failure modes on this kind of data, Claude was consulted to identify candidate approaches from the literature: current yield in place of yield to maturity for bonds in default, and Newey–West standard errors for the autocorrelated weekly price series. The effect of each was checked against the data before it was adopted; for example, yield to maturity gave implausible yields of about 30 percent for one Pennsylvania bond during the state's default. Claude was also used to draft and edit manuscript prose for clarity and length, all of which was reviewed by the author, and flagged a few factual inconsistencies between draft text and source results. The author is responsible for all analytical decisions, verification of method appropriateness, and interpretation of results presented here.

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